

ABILENE, TX, USA

WMO: 722660

Lat: 32.411N

Lon: 99.682W

Elev: 1790

StdP: 13.77

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13962

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	20.1	24.4	4.6	7.4	32.2	8.9	9.3	37.2	29.8	52.8	27.3	53.4	9.8	0	0.558

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	21.2	100.5	70.8	98.4	70.9	96.2	71.0	76.0	90.0	75.0	88.8	74.0	87.6	11.7	160

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
72.4	128.2	80.3	71.4	123.8	79.6	70.4	119.6	79.2	40.6	89.7	39.6	88.6	38.7	87.4	89.4

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
26.3	24.0	20.9	14.1	104.2	3.8	2.9	11.3	106.3	9.0	108.0	6.9	109.6	4.1	111.7		
			12.3	78.2	3.6	2.6	9.7	80.1	7.6	81.6	5.5	83.1	2.9	85.0		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	65.4	45.9	49.6	57.3	65.3	73.4	80.4	83.7	83.5	76.3	66.3	55.3	46.5
	DBStd	16.01	10.37	11.04	10.28	8.73	7.64	4.99	4.29	4.86	6.64	8.71	9.66	9.96
	HDD50	641	205	130	49	6	0	0	0	0	0	6	58	187
	HDD65	2413	595	438	277	102	20	0	0	0	6	91	310	574
	CDD50	6253	76	119	274	465	725	912	1046	1039	790	512	216	79
	CDD65	2551	1	8	37	110	279	462	581	574	346	132	19	2
	CDH74	29088	32	117	369	1107	2911	5306	7313	7051	3555	1149	151	27
	CDH80	14565	3	28	93	397	1320	2699	4036	3912	1666	384	24	3
Wind	WSAvg	10.9	10.8	11.5	12.3	13.2	12.2	11.6	9.9	9.0	9.0	10.4	10.6	10.3
Precipitation	PrecAvg	25.0	1.1	1.2	1.5	1.9	3.0	3.1	2.0	2.8	2.9	2.8	1.5	1.2
	PrecMax	40.5	4.3	3.6	5.2	6.8	9.0	9.6	8.3	8.2	11.0	12.1	5.1	6.3
	PrecMin	13.9	0.0	0.0	0.0	0.0	0.4	0.0	0.0	0.0	0.0	0.1	0.0	0.0
	PrecStd	6.3	1.1	1.0	1.2	1.5	2.0	2.2	2.0	2.2	2.3	2.6	1.3	1.2
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	78.5	84.2	87.5	94.0	100.1	102.2	102.8	103.4	100.0	92.4	83.3	77.7
		MCWB	55.5	57.5	60.2	62.8	68.4	70.7	71.5	70.9	69.8	68.4	61.9	56.0
	2%	DB	71.9	77.3	82.3	88.7	95.0	98.2	99.9	100.3	95.6	87.9	78.4	71.4
		MCWB	53.5	57.1	60.3	63.4	69.1	71.0	71.2	71.0	70.3	67.3	61.4	54.5
	5%	DB	67.0	72.3	78.1	84.6	91.3	95.1	97.8	98.1	92.5	84.3	74.2	67.1
		MCWB	51.9	55.1	59.2	62.9	68.5	71.6	71.5	71.1	69.7	65.4	59.5	53.8
	10%	DB	62.4	67.0	73.6	80.6	87.5	92.2	95.5	95.6	89.3	80.7	70.3	62.7
		MCWB	50.1	53.0	57.4	62.2	68.6	71.8	71.7	71.3	69.4	64.3	57.8	51.5
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	62.2	64.2	67.1	70.8	75.2	77.7	77.4	76.3	75.2	73.0	67.9	63.6
		MCDB	69.0	71.9	77.7	82.2	88.5	91.9	92.1	90.3	87.9	84.6	76.7	68.9
	2%	WB	58.4	61.6	65.0	68.8	73.4	76.1	75.8	75.1	73.8	71.1	65.1	59.4
		MCDB	65.7	69.2	74.5	79.8	85.9	90.0	89.9	89.6	85.6	80.9	72.0	65.6
	5%	WB	54.7	58.8	63.1	67.4	72.2	74.7	74.7	74.2	72.8	69.3	62.6	56.0
		MCDB	62.9	67.7	71.7	77.1	84.3	88.2	88.6	88.7	84.2	78.6	70.0	63.7
	10%	WB	51.1	55.5	60.8	65.6	71.0	73.4	73.6	73.3	71.6	67.8	60.4	52.3
		MCDB	61.6	66.1	69.8	74.6	82.8	86.4	87.4	87.7	82.9	76.6	68.0	61.3
Mean Daily Temperature Range	5% DB	MDBR	23.0	23.3	23.2	24.0	22.1	21.0	21.2	21.3	21.4	22.2	22.1	22.0
		MCDBR	28.3	29.2	27.6	27.7	26.4	24.3	23.6	23.6	24.3	25.4	24.9	27.2
		MCWBR	15.3	15.6	13.4	12.9	9.7	7.8	7.1	7.0	7.7	11.1	12.8	15.3
	5% WB	MCDBR	24.0	25.1	21.8	22.5	22.4	21.0	20.8	20.7	20.0	20.4	20.7	21.5
MCWBR		15.6	16.6	12.2	12.7	9.9	8.3	7.0	6.9	7.4	10.3	13.3	14.9	
Clear-Sky Solar Irradiance	taub	0.289	0.303	0.324	0.367	0.391	0.406	0.416	0.404	0.374	0.350	0.317	0.296	
	taud	2.544	2.508	2.468	2.358	2.359	2.349	2.381	2.397	2.420	2.472	2.493	2.528	
	Ebn at Noon	299	304	304	292	285	279	275	278	283	283	285	289	
	Edh at Noon	26	30	34	39	40	40	38	37	35	31	27	25	
All-Sky Solar Radiation	RadAvg	1013	1223	1601	1946	2042	2246	2201	2008	1683	1350	1066	890	
	RadStd	108	167	112	150	199	171	144	140	129	200	127	103	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	+0.58	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+223	+160
(47)	Regional (18 neighbors)	+0.59	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+227	+151

Nomenclature: See separate page